

Gesture Control Bluetooth Car 🚗 🖐️

Developer: THIRUPATHI AASHRITH

Core Technologies: Arduino, ADXL345 Accelerometer, HC-05 Bluetooth, L298N H-Bridge Motor Driver

A real-time, gesture-controlled robotic vehicle powered by **Arduino Uno**, an **ADXL345 3-Axis Accelerometer**, **L298N Motor Driver**, and paired **HC-05 Bluetooth Modules**.

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🎯 Project Overview & Purpose

Designed and developed by **THIRUPATHI AASHRITH**, this project replaces traditional joysticks or smartphone interfaces with intuitive hand gesture controls to pilot a 2WD/4WD robotic chassis.

- **The Glove (Transmitter):** Reads hand pitch and roll using an ADXL345 accelerometer, converts tilt data into movement commands, and broadcasts them via an HC-05 Bluetooth module operating in **Master Mode**.
- **The Car (Receiver):** Receives incoming gesture commands via a secondary HC-05 module operating in **Slave Mode**, processing them on an Arduino Uno to direct an L298N motor driver using Pulse Width Modulation (PWM) speed control.

🌍 Real-World Applications

Gesture recognition technology bridges physical hardware interaction with natural human movement. Real-world applications include:

- **Search and Rescue Operations:** Enables remote operators to maneuver hazard-entry rovers without looking down at complex controllers.
- **Assistive Mobility:** Provides intuitive control options for electric wheelchairs for individuals with limited finger dexterity.
- **Hazardous Material Handling:** Allows technicians to pilot robotic arms or vehicles while wearing bulky protective gloves.
- **Industrial Automation:** Hands-free control over automated guided vehicles (AGVs) in smart warehouse settings.

Hardware Components

Component	Quantity	Purpose
Arduino Uno R3	2	Primary microcontrollers for Glove and Car
ADXL345 Accelerometer	1	Measures 3-axis hand tilt on the gesture glove
HC-05 Bluetooth Modules	2	Wireless Master/Slave serial bridge
L298N Dual H-Bridge Motor Driver	1	Controls motor direction and speed
2WD or 4WD Robot Chassis Kit	1	Frame with DC motors and wheels
7.4V / 11.1V Li-Ion / LiPo Battery	1	High-current power source for motors
10k Ω Resistors	6	Forms voltage dividers to step 5V signals down to 3.3V
Breadboards & Jumper Wires	--	Circuit connections



Circuit & Wiring Details

1. Transmitter Unit (Glove)

Note: SoftwareSerial uses **Pin 2 (RX)** and **Pin 3 (TX)** to leave Pins 0 & 1 free for USB programming.

- **ADXL345 Accelerometer Wiring:**

- **VCC** => Arduino **3.3V**
- **GND** => Arduino **GND**
- **SDA** => Arduino **A4** (I2C Data)
- **SCL** => Arduino **A5** (I2C Clock)
- **CS** => Arduino **3.3V** (Forces I2C mode)
- **SDO** => Arduino **GND** (Sets address to **0x53**)

- **HC-05 Bluetooth (Master) Wiring:**

- **VCC** => Arduino **5V**
- **GND** => Arduino **GND**
- **TXD** => Arduino **Pin 2**
- **RXD** => Connected through a **Voltage Divider** from Arduino **Pin 3** (10kΩ to Pin 3, 20kΩ to GND).

2. Receiver Unit (Car)

- **HC-05 Bluetooth (Slave) Wiring:**

- **VCC** => Arduino **5V**
- **GND** => Arduino **GND**
- **TXD** => Arduino **Pin 2**
- **RXD** => Connected through a **Voltage Divider** from Arduino **Pin 3** (10kΩ to Pin 3, 20kΩ to GND).

- **L298N Motor Driver Wiring:**

- **Jumpers on ENA / ENB must be removed!**
- **ENA** => Arduino **Pin 9** (PWM Speed - Left Motor)
- **IN1** => Arduino **Pin 4** (Left Motor Direction)
- **IN2** => Arduino **Pin 5** (Left Motor Direction)
- **IN3** => Arduino **Pin 6** (Right Motor Direction)
- **IN4** => Arduino **Pin 7** (Right Motor Direction)
- **ENB** => Arduino **Pin 10** (PWM Speed - Right Motor)
- **12V Terminal** => Battery Positive (+)
- **GND Terminal** => Battery Negative (-) **AND Arduino GND (Crucial Common Ground)**
- **5V Output** => Arduino **5V Pin** (Powers the Arduino from the main battery)

Bluetooth Pairing Guide (HC-05 AT Commands)

Before running the main code, both HC-05 modules must be configured via AT Commands to automatically pair with each other on startup.

Step 1: Configuration Mode Entry

1. Disconnect power from the HC-05 module.
2. Press and hold the small onboard push-button while re-connecting power.
3. The LED will blink slowly (once every 2 seconds), indicating **AT Command Mode**.

Step 2: Configure Slave Unit (Car Side)

1. Set Serial Monitor to **38400 Baud Rate** and **"Both NL & CR"**.
2. Send **AT** (Verify response: **OK**).
3. Send **AT+ROLE=0** to configure as **Slave**.
4. Send **AT+ADDR?** to retrieve the device address (e.g., **+ADDR:98d3:31:f62a40**).
5. **Note down this address** for the Master unit setup.

Step 3: Configure Master Unit (Glove Side)

1. Boot the glove's HC-05 into AT Mode using the same method.
2. Send **AT** (Verify response: **OK**).
3. Send **AT+ROLE=1** to configure as **Master**.
4. Send **AT+CMODE=0** to restrict connections exclusively to a bound address.
5. Send **AT+BIND=98d3,31,f62a40** (Replacing colons in the Slave address with commas).

Once configured, both modules will automatically establish a secure link upon boot (indicated by a synchronized double-blink sequence).

System Architecture & Logic

1. Gesture Detection Mapping

The ADXL345 evaluates acceleration along the X and Y axes in g -force vectors (m/s^2):

- **Forward ('F'):** Hand tilted downward along the positive X-axis ($X > 3.5$).
- **Backward ('B'):** Hand tilted upward along the negative X-axis ($X < -3.5$).
- **Left ('L'):** Hand rolled to the left along the positive Y-axis ($Y > 3.5$).
- **Right ('R'):** Hand rolled to the right along the negative Y-axis ($Y < -3.5$).
- **Stop ('S'):** Hand maintained flat in the resting deadzone ($-3.5 \leq X, Y \leq 3.5$).

2. Motor Control Logic

The L298N driver receives directional signals (**IN1–IN4**) and Pulse Width Modulation (PWM) speed signals (**ENA, ENB** mapped to values from **0** to **255**):

- **Forward:** Both left and right motor banks rotate forward.
- **Backward:** Both left and right motor banks rotate in reverse.
- **Left Turn:** Left motor bank reverses while the right motor bank drives forward (Differential Pivot Turn).
- **Right Turn:** Left motor bank drives forward while the right motor bank reverses (Differential Pivot Turn).
- **Stop:** All directional pins set to **LOW** and PWM outputs set to **0**.



Installation & Setup Guide

1. **Hardware Assembly:** Wire the Transmitter and Receiver circuits according to the pinout details provided in Section 4.
2. **Library Installation:** Open Arduino IDE, navigate to [Sketch > Include Library > Manage Libraries...](#), search for [ADXL345](#) and install the **Adafruit ADXL345** library along with its dependencies ([Adafruit Unified Sensor](#)).
3. **Bluetooth Pairing:** Execute the AT command process detailed in Section 5.
4. **Upload Transmitter Code:** Open [Glove_Transmitter.ino](#) and flash it to the glove Arduino Uno.
5. **Upload Receiver Code:** Open [Car_Receiver.ino](#) and flash it to the car Arduino Uno.
6. **Execution:** Power the car chassis via its high-current battery pack, then power the gesture glove. Once the Bluetooth modules connect (double-blink indicator), tilt your hand to pilot the car!



Troubleshooting Checklist

- **ADXL345 Returns "No Sensor Found":** Run an I2C Scanner sketch. Verify that [CS](#) is connected to [3.3V](#), [SDO](#) is connected to [GND](#), and the I2C address returns [0x53](#).
- **Bluetooth Packets Received, But Motors Don't Turn:** Ensure the **L298N GND and Arduino GND are connected**. Check that the battery supply outputs at least **7.4V** under motor load.
- **Car Turns Opposite of Gesture:** Reverse the two motor leads in the L298N terminal block for the affected side, or flip the sensor orientation on the glove.



Author & Credits

Project Creator & Lead Engineer: THIRUPATHI AASHRITH

- Created for robotics, embedded systems, and human-computer interaction (HCI) exploration.
- Open-source and free for educational use.